

$$(1) \Rightarrow [M][\phi] \ddot{\vec{q}}(t) + [C][\phi] \dot{\vec{q}}(t) + [K][\phi] \vec{q}(t) = \vec{f}(t) \quad (1)$$

we left multiply by $[\phi]^T$:

$$\Rightarrow \underbrace{[\phi]^T [M] [\phi]}_{[\tilde{M}]} \ddot{\vec{q}}(t) + \underbrace{[\phi]^T [C] [\phi]}_{[\tilde{C}]} \dot{\vec{q}}(t) + \underbrace{[\phi]^T [K] [\phi]}_{[\tilde{K}]} \vec{q}(t) = \underbrace{[\phi]^T \vec{f}(t)}_{\vec{\tilde{f}}(t)}$$

$$\Rightarrow \boxed{[\tilde{M}] \ddot{\vec{q}}(t) + [\tilde{C}] \dot{\vec{q}}(t) + [\tilde{K}] \vec{q}(t) = \vec{\tilde{f}}(t)} \quad \boxed{\vec{\tilde{f}}(t) = [\phi]^T \vec{f}(t)}$$

$\downarrow$ diagonal $\downarrow$ diagonal? No $\downarrow$ diagonal Vector of generalized forces

To diagonalise $[C]$ too, we need additional assumptions of the damping matrix $[C]$:

* diagonal damping:

we forget $[C]$ and we directly assume that $[\tilde{C}]$ is diagonal.

$$[\tilde{C}] = \begin{bmatrix} \tilde{c}_1 & & (0) \\ & \ddots & \\ (0) & & \tilde{c}_n \end{bmatrix}$$

How to calculate $\tilde{c}_1, \tilde{c}_2, \dots, \tilde{c}_n$.

We set values or identify experimentally $\tilde{\xi}_1, \tilde{\xi}_2, \dots, \tilde{\xi}_n$

$$\left(\xi = \frac{c}{2\sqrt{mk}} \right)$$

$$\text{Then } \begin{cases} \tilde{c}_1 = 2 \tilde{\xi}_1 \sqrt{\tilde{m}_1 \tilde{k}_1} \\ \vdots \\ \tilde{c}_n = 2 \tilde{\xi}_n \sqrt{\tilde{m}_n \tilde{k}_n} \end{cases}$$

* Rayleigh damping: we assume

$$[C] = a[M] + b[K] \quad (a, b) \in \mathbb{R}^2$$

$$\Rightarrow [\tilde{C}] = [\phi]^T [C] [\phi] = a [\phi]^T [M] [\phi] + b [\phi]^T [K] [\phi] \\ = a [\tilde{M}] + b [\tilde{K}] \rightarrow \text{diagonal}$$

(2)

$$= \begin{pmatrix} a\tilde{m}_1 + b\tilde{k}_1 & (0) \\ (0) & a\tilde{m}_n + b\tilde{k}_n \end{pmatrix}$$

$$= \begin{pmatrix} \tilde{c}_1 & (0) \\ (0) & \tilde{c}_n \end{pmatrix}$$

How to calculate a and b ?

Again, we use information on $\tilde{\xi}_1, \dots, \tilde{\xi}_n$

$$\Rightarrow \begin{cases} \tilde{c}_1 = a\tilde{m}_1 + b\tilde{k}_1 = 2\tilde{\xi}_1 \overline{\tilde{m}_1 \tilde{k}_1} \\ \vdots \\ \tilde{c}_n = a\tilde{m}_n + b\tilde{k}_n = 2\tilde{\xi}_n \overline{\tilde{m}_n \tilde{k}_n} \end{cases}$$

we have n equations to calculate a and b .

$\Rightarrow$ we keep only two equations (in general, the first one and the last one, and we calculate a and b).

$\Rightarrow$ we can't control all the damping ratios with the Rayleigh model.

• We have the following equations

(3)

$$\begin{cases} [M] \ddot{\vec{x}} + [C] \dot{\vec{x}} + [K] \vec{x} = \vec{f}(t) & (1) \\ \vec{x}(0) = \vec{x}_0 & \dot{\vec{x}}(0) = \vec{v}_0 \end{cases}$$

We introduce the following change of variable:

$$\boxed{\vec{x}(t) = [\Phi] \vec{q}(t) = \sum_{\alpha=1}^n q_{\alpha}(t) \vec{\Phi}_{\alpha}}$$

$$\vec{q}(t) = \begin{pmatrix} q_1(t) \\ q_2(t) \\ \vdots \\ q_n(t) \end{pmatrix} : \text{vector of the generalized coordinates}$$

Interpretation:

• Before : $\vec{x}(t) = \begin{pmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{pmatrix} = x_1(t) \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + x_2(t) \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} + \dots + x_n(t) \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$

• Now : $\vec{x}(t) = q_1(t) \vec{\Phi}_1 + q_2(t) \vec{\Phi}_2 + \dots + q_n(t) \vec{\Phi}_n$

$q_1(t), q_2(t), \dots, q_n(t)$ are the ~~ampli~~ variations of oscillations of the modes $\vec{\Phi}_1, \vec{\Phi}_2, \dots, \vec{\Phi}_n$

$\vec{x}(t)$ is rewritten as a combination of the elastic modes with varying amplitudes $\rightarrow$ modal superposition.

Advantage: the modes are independent. (but $x_1, x_2, \dots, x_n$ are not)

$$\vec{x}(t) = [\Phi] \vec{q}(t) \Rightarrow \begin{cases} \dot{\vec{x}}(t) = [\Phi] \dot{\vec{q}}(t) \\ \ddot{\vec{x}}(t) = [\Phi] \ddot{\vec{q}}(t) \end{cases}$$

Modal equations:

(4)

$$[\tilde{M}] \ddot{\vec{q}}(t) + [\tilde{C}] \dot{\vec{q}}(t) + [\tilde{K}] \vec{q}(t) = \vec{\tilde{f}}(t)$$

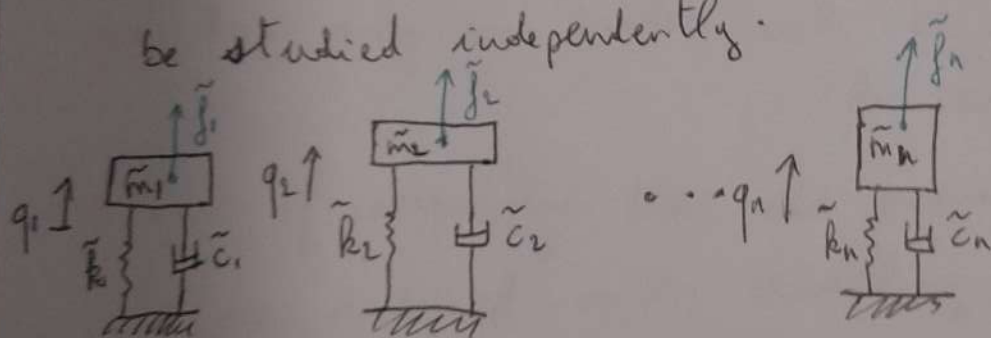
$$\hookrightarrow \begin{pmatrix} \tilde{m}_1 & (0) \\ (0) & \tilde{m}_n \end{pmatrix} \begin{pmatrix} \ddot{q}_1 \\ \vdots \\ \ddot{q}_n \end{pmatrix} + \begin{pmatrix} \tilde{c}_1 & (0) \\ (0) & \tilde{c}_n \end{pmatrix} \begin{pmatrix} \dot{q}_1 \\ \vdots \\ \dot{q}_n \end{pmatrix} + \begin{pmatrix} \tilde{k}_1 & (0) \\ (0) & \tilde{k}_n \end{pmatrix} \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix} = \begin{pmatrix} \tilde{f}_1 \\ \vdots \\ \tilde{f}_n \end{pmatrix}$$

Now, all the matrices are diagonal.

$\hookrightarrow$ we have now n uncoupled equations:
independent

$$\begin{cases} \tilde{m}_1 \ddot{q}_1 + \tilde{c}_1 \dot{q}_1 + \tilde{k}_1 q_1 = \tilde{f}_1 \\ \vdots \\ \tilde{m}_n \ddot{q}_n + \tilde{c}_n \dot{q}_n + \tilde{k}_n q_n = \tilde{f}_n \end{cases}$$

$\Rightarrow$ we obtain n independent oscillator equations, which can be studied independently.



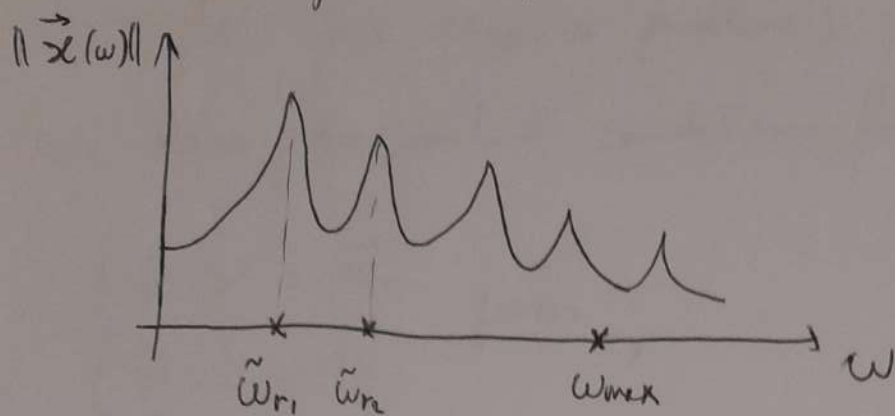
$\Rightarrow$ All the results established for an oscillator (lectures 2, 3 and 4) can be used to calculate and study $q_1(t)$, $q_2(t)$, ..., $q_n(t)$.

Once $q_1(t)$, $q_2(t)$, ..., $q_n(t)$ are calculated, we go back to physical coordinates:

$$\begin{aligned} \vec{x}(t) &= [\phi] \vec{q}(t) \\ &= \sum_{\alpha=1}^n \phi_{\alpha}(t) \vec{p}_{\alpha} \end{aligned}$$

Ex. Harmonic forced response:

⑤



pb : for a real-life structure $n > 1000000$
 $\Rightarrow$ more than 1000000 modes

modal truncation

We only select the modes for which the frequency are in the frequency range of interest $[0, w_{max}]$.

More precisely, all the modes such that $w_n < 1.2 w_{max}$ we keep only m modes, with $m \ll n$

(for instance $m = 100$)

$\Rightarrow$ Save of computation time and memory requirement.

$\Rightarrow$ reduced-order modelling.

$$\vec{x}(t) = \sum_{\alpha=1}^m q_{\alpha}(t) \vec{\phi}_{\alpha}$$

$$= [\Phi^t] \vec{q}(t)$$

with $[\Phi^t] = [\vec{\phi}_1, \vec{\phi}_2, \dots, \vec{\phi}_m, \vec{\phi}_{m+1}, \dots, \vec{\phi}_n]$

$\downarrow$
 neglected

- Initial conditions for the $q_1, q_2, \dots, q_n$?
(for free response problems).

⑥

We know the initial conditions for $\vec{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}$

$$\begin{cases} \vec{x}(0) = \vec{x}_0 \\ \dot{\vec{x}}(0) = \vec{v}_0 \end{cases} \text{ given}$$

$$\Rightarrow \begin{cases} [\Phi] \vec{q}(0) = \vec{x}_0 \\ [\Phi] \dot{\vec{q}}(0) = \vec{v}_0 \end{cases}$$

we left multiply by $[\Phi]^T [M]$:

$$\Rightarrow [\Phi]^T [M] [\Phi] \vec{q}(0) = [\Phi]^T [M] \vec{x}_0$$

$$[\Phi]^T [M] [\Phi] \dot{\vec{q}}(0) = [\Phi]^T [M] \vec{v}_0$$

$[\tilde{M}]$

$\Rightarrow$

$$\begin{aligned} \vec{q}(0) &= [\tilde{M}]^{-1} [\Phi]^T [M] \vec{x}_0 \\ \dot{\vec{q}}(0) &= [\tilde{M}]^{-1} [\Phi]^T [M] \vec{v}_0 \end{aligned}$$